A Study On Integrated Energy System Long-term Planning Optimization Considering Complementarity And Economy

Nan Zhang
Polytechnic Institute
Zhejiang University
Hangzhou, China
nan_zhang@zju.edu.cn

Xiaojie Lin*
College of Energy Engineering
Zhejiang University
Hangzhou, China
xiaojie.lin@zju.edu.cn

Wei Zhong
College of Energy Engineering
Zhejiang University
Hangzhou, China
zhongw@zju.edu.cn

Fanqi Kong
College of Energy Engineering
Zhejiang University
Hangzhou, China
3170103678@zju.edu.cn

***Abstract*—Carbon reduction and neutrality receive more and more attention due to concerns about global warming and the energy crisis. The access of renewable energy brings greater volatility and uncertainty to the integrated energy system (IES). Analyzing the relationship between complementarity and economy is the key to maintaining the long-term benefits of IES. To illustrate the multi-energy coordination and complementary effect in IES, a two-stage planning optimization method considering economy and complementarity is proposed in this paper. Firstly, the complementarity quantification index is established by analyzing the correlation of different energy net loads. Meanwhile, the comprehensive economic index is formed considering the system construction, operation, and maintenance. Then, the energy system is modeled, and the device configuration simulation is carried out. Finally, the Pareto solution set of the programming scheme is obtained by the NSGA-II algorithm. Measured by proposed indicators, the annual total costs increase by 15.8% when the system complementarity is enhanced by 41.3%.**

Keywords—complementarity, integrated energy system, two-stage planning

I. INTRODUCTION

Considering the "carbon peaking and carbon neutrality goals", the proportion of renewable energy in the integrated energy system (IES) will continue to increase in China. Due to the strong fluctuation and intermittence of new energy power generation, the effective quantification and utilization of complementary effects between various energy carriers will be a key to the safety and stability of the energy system. Therefore, the quantitative analysis of economics and complementarity is necessary for long-term IES planning optimization.

To evaluate the complementarity effect of comprehensive energy, partial scholars put forward effective indicators, mainly including correlation coefficient and output volatility. Correlation coefficients such as Pearson correlation coefficient (PCCs)[1], Kendall correlation coefficient (KCCs)[2], and Spearman correlation coefficient (SCCs)[3] can quantify the degree of correlation between time series, which supports continuous explorations to measure complementarity of IES. Different correlation coefficients are suitable for different scenes. PCCs are ideal for continuous sequences that meet normal distribution, such as wind speed, solar radiation, and load data, with a relatively simple calculation. KCCs are suitable for discrete ordered variables that do not meet normal distribution, which must be sorted once, with high time complexity. SCCs are suitable for continuous-time series that do not meet normal distribution, which is required to be sorted twice, with the highest time complexity.

Output volatility is generally defined as the difference in power output[4] between the devices in two adjacent moments[5]. Due to the excessive variety of equipment, some scholars prefer to measure the complementarity through the fluctuation or reliability of the whole system directly[6]. Taking a wind-solar-hydropower system as a typical example, Beluco et al.[7] started from three perspectives: peak-valley time interval, the proximity of accumulative power value, and amplitude similarity, while Borba et al.[8] considered the stability of output, the proportion of undersupply and Han et al.[9] considered the short-term, long-term fluctuation of output. Output volatility of a system is the synthesis of devices, in essence, which is superior for the complex energy system.

The planning and design of a multi-energy flow system are further carried out on the basis of complementarity and traditional economic index studies. Given the new energy power generation, nearly all of the studies on complementarity take electricity as their core flow, and only scanty studies about CHP take the heat as the core energy flow[10]. Most scholars choose economics as the ultimate goal, as the traditional planning method. Complementarity indicators are used to measure the characteristics of multiple energy[11] in IES to establish the demand-side model that allows for response with the minimum dispatching cost. On the source side[12], it is used to evaluate the synergy characteristics of random energy such as wind and solar to ensure the economic optimality of distribution network construction as the ultimate goal. Meanwhile, Ye et al.[13] research the advantage of the wind-solar-hydropower system compared with the single energy supply system in terms of load tracking capability through complementarity indicators. In general, few studies consider multiple economic and complementary indicators in IES planning.

II. MODODELING METHODOLOGY

A. Quantitative method of complementarity

The power output fluctuation of the energy supply equipment is defined as the equipment output difference between two adjacent moments, as shown in Equations (1) and (2).

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$$FR = \frac{1}{n-1}\sum_{t=1}^{T-1}|\gamma(t)| \quad (1)$$

$$\gamma(t) = \frac{P(t+1)-P(t)}{P_{cap}}, t = 1,2,\ldots, \quad T-1 \quad (2)$$

Where, FR is the fluctuation rate of equipment output time series; $(P_1, P_2, \ldots, P_n)$ is the output value of the equipment at time t (kW); P_{cap} is the rated power of the equipment (kW); $\gamma(t)$ is the fluctuation rate of the t interval in the equipment output time series.

The complementarity index established on this basis is shown in Equation (3)-(6).

$$CROF = 1 - \frac{FROC}{FROI} \quad (3)$$

$$FROI = \frac{\alpha_1 FR_1 + \alpha_2 FR_2 + \cdots + \alpha_k FR_k}{\alpha_1 + \alpha_2 + \cdots + \alpha_k} \quad (4)$$

$$FROC = \frac{1}{n-1}\sum_{t=1}^{T-1}|\gamma^c(t)| \quad (5)$$

$$\gamma^c(t) = \frac{\alpha_1\gamma_1(t) + \alpha_2\gamma_2(t) + \cdots + \alpha_k\gamma_k(t)}{\alpha_1 + \alpha_2 + \cdots + \alpha_k} \quad (6)$$

Where, $CROF$ is the complementarity index of the multi-energy system; α_k is the rated power share ratio of the k device in the system; $FROI$ is the weighted average of fluctuation when the power supply equipment operates alone; $FROC$ is the overall output fluctuation rate; $\gamma^c(t)$ is the output fluctuation rate of in time period T; $\gamma_k(t)$ is the output fluctuation rate of equipment k in time period t.

Considering the peak-valley characteristics of user load, this paper adopts the fluctuation rate of net load to measure the complementarity of energy system[11]. The lower the fluctuation rate indicates that the output can better conform to the peak-valley characteristics of regional energy consumption. In this case, only stable additional electric energy from the external power grid is needed to satisfy the energy consumption of the system, thus reducing the impact on the external power grid.

The net load of the system is defined as the difference between the overall system output and the regional load. Then the fluctuation of the net load series is described by the standard deviation. Finally, the fluctuation of the net load of different energy is added. The smaller the value of complementarity index is, the more the system output conforms to the user-side energy consumption mode, and the stronger the complementarity of the system is, as shown in Equatitions (7)-(9).

$$CONL = \sum_{i\in En} std(NL_i) = \sum_{i\in En}\sqrt{\frac{\sum_{t=1}^{T}(NL_i(t)-\overline{NL_i})^2}{T-1}} \quad (7)$$

$$NL_i = (NL_i(1), NL_i(2), \ldots, NL_i(t), \ldots, NL_i(T)) \quad (8)$$

$$NL_i(t) = D_i(t) - \sum_{j\in Eq} P_j(t) \quad (9)$$

Where,$CONL$ is the complementarity indicator of the IES (kW); NL_i is the time series of net load for different energy flows i (such as electricity, heat, cold); $NL_i(t)$ and $\overline{NL_i}$ are the net load value and mean value (kW) of net load time series at time t respectively; En is the set of symbols for different energy flows; $D_i$is the regional load demand(kW) of energy flows i; P_j is the output power (kW) of energy device j.

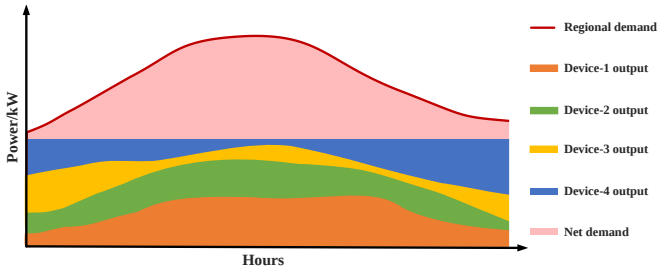

Fig. 1. Diagram of IES with weak complementary

To illustrate the advantages of the quantitative method above, the device output of IES shown in Fig. 1 is analyzed. In the comprehensive energy system with faint complementary, the peak-valley features of regional demand are entirely reflected in the net demand power. So the external grid has to bear the fluctuation of regional demand entirely, suffering enormous peak regulation pressure. However, if the correlation coefficient or output volatility is directly used for evaluation, it is noteworthy but unreasonable that the calculation result is 0 and indicate that the system complementarity has reached the strongest state.

B. *Quantitative method of Economy*

A long-term planning of IES is supposed to consider various aspects of design and operation, including the initial investment cost, operation cost, and maintenance cost, and the objective function of total cost is shown in Equation (10).

$$TC = IC + OC + MC \quad (10)$$

Where, Obj_{upper} is the objective function value of the upper devices allocation planning model (€/year); TC,IC,OC,$MC$are respectively the annual total cost, annual investment cost, annual operation cost, and annual maintenance cost (€/year) of the energy system, and the annual investment cost includes the initial investment cost of all devices by years, as shown in Equation (11).

$$IC = \sum_{i\in Eq} Co_i \cdot Pe_i \cdot \frac{IR(1+IR)^{l_i}}{(1+IR)^{l_i}-1} \quad (11)$$

Where,Co_i and Pe_i are the initial investment cost(€/kW) and rated power (kW) of device i respectively; IR is the investment discount rate (%); l_i is the service life (years) of device i; Eq a set of symbols for all types of devices.

The maintenance cost of the unit can be converted into the initial investment cost in accordance with a certain proportion so that the annual maintenance cost can be expressed as Equation (12).

$$MC = \sum_{i\in E} \zeta_i \cdot Co_i \cdot Pe_i \cdot \frac{IR(1+IR)^{l_i}}{(1+IR)^{l_i}-1} \quad (12)$$

Where, $\zeta_i$is the maintenance coefficient of device i.

The objective function of operating cost is shown in Equation (13).

$$OC = \sum_{t=1}^{T}\sum_{i\in En} Co_i^{buy}(t) \cdot P_i^{buy}(t) \cdot \Delta t - \sum_{t=1}^{T}\sum_{i\in En} Co_i^{sell}(t) \cdot P_i^{sell}(t) \cdot \Delta t + Co_{cap} \cdot max(P_{electricity}^{buy}(t)) \quad (13)$$

Where, Co_i^{buy} and$P_i^{buy}$are the price (€/kWh) and power (kW) of energy flow i purchased from the external network respectively; Co_i^{sell} and$P_i^{sell}$are the price (€ /kWh) and power (kW) of energy flow i sold to the external network,

respectively; $En$is the symbol set of different types of energy flows. Δt is the time step of operating; Co_{cap} is the capacity fee of purchasing electricity (€/kW).

C. Two-stage optimization method

Multi-objective optimization of IES is carried out based on economy and complementarity, and the planning problem is divided into two parts by a two-stage optimization planning process. The upper model optimizes the capacity allocation of various candidate devices and transmits different planning schemes to the lower model. The lower model optimizes the operation and returns the indicators results to the upper model. The upper model aims at economy and complementarity, using a multi-objective optimization algorithm but not a deterministic optimization algorithm. The lower model calculates not only the economic results but also the complementarity under the current planning scheme. Its framework is shown in Fig. 2.

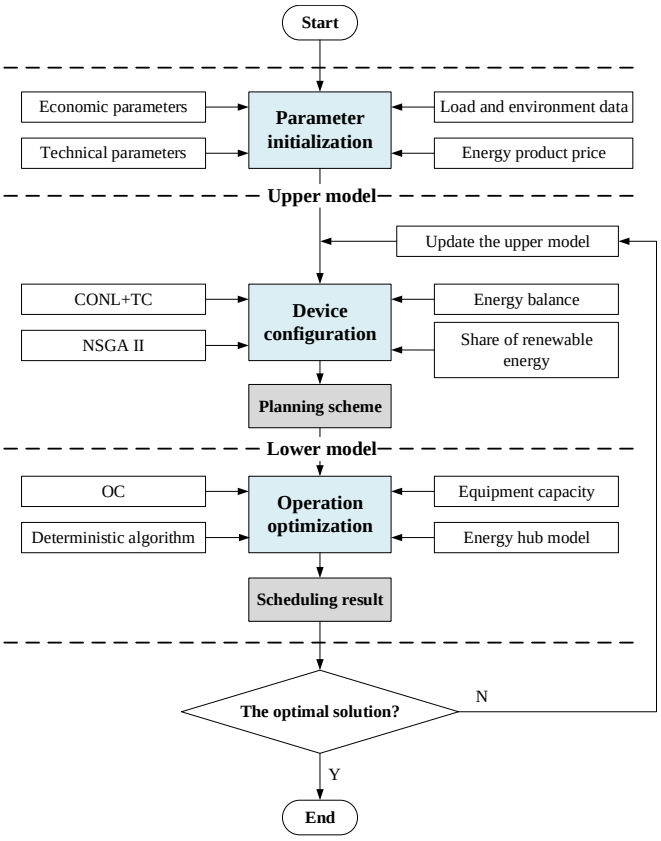

Fig. 2. Framework of two-stage planning method based on economy and complementarity

For capacity allocation, economy and complementarity are taken as the fitness function of each individual in the parent and child population. Non-dominated sorting genetic algorithm II (NSGA- II) is used to solve the multi-objective problem and generate a set of solutions, often called Pareto solutions. The specific steps are as follows:

- Set up the boundary conditions and initials parameters, including economy and technical parameters of the candidate device, load data of typical-day (such as electricity, heat, cold demand), meteorological data (such as air temperature, solar radiation, and wind speed), energy commodity prices.
- Start the heuristic multi-objective optimization algorithm to solve the upper model. According to the economic and complementary objective function value, numerous planning schemes (the number is determined by the population size) can be generated with the constraint of energy balance and the renewable energy proportion.
- Input the planning schemes into the lower operation model. Under the load and environmental parameters of typical days, the capacity limit of devices determined above is taken into account and the deterministic algorithm is used to get the operation scheduling results corresponding to each planning scheme.
- Judge whether the optimal set of planning schemes has been obtained (depending on the termination condition set by the upper model). If so, output all planning schemes and end process; If not, update the economic function value in the planning scheme with the cost in the operation scheduling result, and update the complementary function value with the device output, then repeat step 2).

III. Case Study

A. Energy System Description

The structure of the IES for a planning case study is shown in Fig. 3, which consists of three parts: producer, converter, and consumer. There are a variety of devices, such as gas turbines, absorption chiller units, and storage devices, as well as renewable energy sources, including wind turbines and photovoltaics. In addition, it is connected to the external energy commodity to ensure the energy balance when running out of capacity.

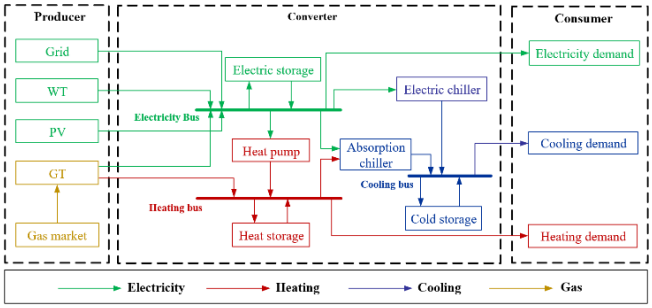

Fig. 3. The energy supply structure of IES

B. Data Collection

The electricity price in this IES reference a two-part annual electricity price provided by the power grid operator[14]. The charging standard is 114.29 €/(kW·a) when the yearly electricity consumption time is in excess of 2,500 hours. The capacity is determined according to the annual maximum demand, and the standard electricity price is 0.0025 €/kWh. The natural gas price reference the market monitoring report issued by the government[15], taken as 0.0286 €/kWh constantly. Moreover, the discount rate of energy sold by the case is 8%, and the peak load is regarded as the minimum capacity allocation of renewable energy output, given the increasing utilization and intermittent production of renewable energy. The meteorological data reference NASA Merra-2 database[16], which provides a spatial resolution of 0.5°×0.625° and has covered meteorological data since 1980. The energy demand data time series reference energy consumption data set published by Scientific Data[17], as shown in Fig. 4.

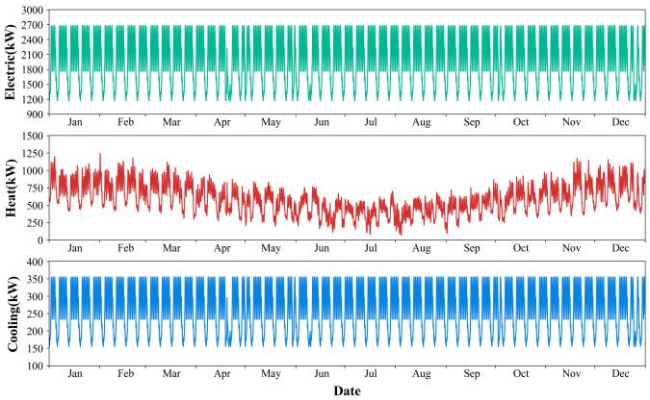

Fig. 4. Annual load data of electric, heat, and cooling

C. Devices Modeling

The equipment in the IES can be roughly divided into two categories: energy conversion equipment and energy storage equipment. Based on the Energy hub modeling method, the devices above case can be established as follows:

1) Energy Conversion

This paper pays less attention to the physical modeling process of various energy flow production equipment but focuses on its modularization consideration and describes its output as:

$$\mu_{h,i}^{E}(t)P_{h,min,i}^{E} \le P_{h,i}^{E}(t) \le \mu_{h,i}^{E}(t)P_{h,max,i}^{E} \tag{14}$$

$$-P_{h\Delta t,max,i}^{E} \le P_{h,i}^{E}(t) - P_{h,i}^{E}(t-1) \le P_{h\Delta t,max,i}^{E} \tag{15}$$

$$\mu_{h,i}^{E}(t) = \{0,1\} \tag{16}$$

Among them: $P_{h,i}^{E}(t)$ is the output of energy supply equipment i at the moment t; $P_{h,min,i}^{E}$ and $P_{h,max,i}^{E}$ are the minimum and maximum outputs of energy supply equipment i; $\mu_{h,i}^{E}(t)$ is the start-stop state of device i at the moment t; $P_{h\Delta t,max,i}^{E}$ is the output variation limit of device i in a scheduling period Δt.

2) Energy Storage

Energy storage equipment is the guarantee of flexible regulation of comprehensive energy stations. This paper describes the model of system heat storage equipment as follows:

$$\begin{cases} S_h^F(t) = \eta_h^F S_h^F(t-1) + \eta_{\mathrm{hi}}^F Q_{\mathrm{hi}}^F(t) - Q_{\mathrm{ho}}^F(t)/\eta_{\mathrm{ho}}^F \\ 0 \le Q_{\mathrm{hi}}^F(t) \le Q_{\mathrm{hi,max}}^F \\ 0 \le Q_{\mathrm{ho}}^F(t) \le Q_{\mathrm{ho,max}}^F \\ 0 \le \mu_{\mathrm{hi}}^F + \mu_{\mathrm{ho}}^F \le 1 \\ 0 \le S_h^F(t) \le S_{\mathrm{h,max}}^F \end{cases} \tag{17}$$

Among them: $S_h^F(t)$ is the residual heat of heat storage equipment at the moment t; η_h^F is the energy residual rate of heat storage equipment after unit time; $Q_{\mathrm{hi}}^F(t)$, $Q_{\mathrm{ho}}^F(t)$ are respectively the heat storage and release the power of the heat storage equipment at the time; η_{hi}^F、η_{ho}^F are respectively the energy storage efficiency and discharge efficiency of equipment; $Q_{\mathrm{hi,max}}^F$、$Q_{\mathrm{ho,max}}^F$ are respectively the maximum heat storage power and heat release power of heat storage equipment; μ_{hi}^F、μ_{ho}^F are respectively the heat release state of energy storage equipment; $S_{\mathrm{h,max}}^F$ is the maximum heat storage of the heat storage device.

IV. Results

IES in this case is planned with a designed service life of 20 years. A Pareto solution set is obtained through the proposed two-stage multi-objective optimization planning method, containing 50 planning schemes shown in Fig. 5.

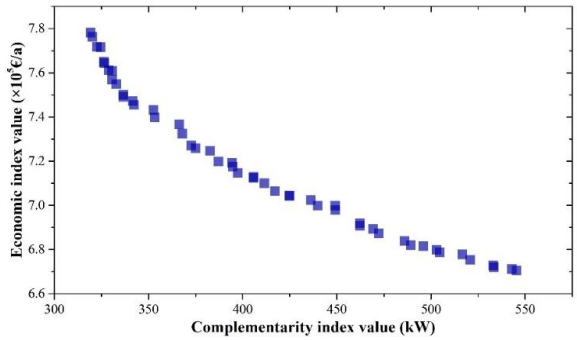

Fig. 5. Economy and complementarity index of planning schemes in the Pareto solution set

It is seen that the economy and complementarity of IES are contradictory, while the increase of complementarity requires higher economic costs. Furthermore, the slope of the economy-complementarity curve decreases with the rise of the complementarity value, indicating that the marginal cost of improving complementarity is small when the annual cost is negligible.

To emphasize the characteristics of various planning schemes, a subset of all the Pareto-optimal solutions is selected after eliminating the results with high coincidence, containing 25 planning schemes. The relationships of complementary and economic indicators of the selected planning scheme are shown in Fig. 6. Investment and maintenance costs are determined after the planning scheme is finalized and have nothing to do with the subsequent operation.

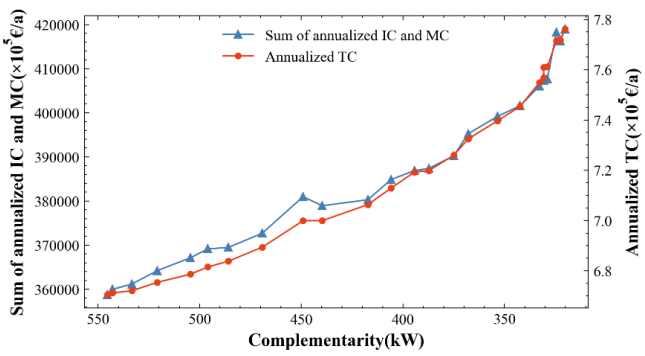

Fig. 6. The relationships of complementarity and economic indicators of the selected planning scheme

TABLE I. Table Type Styles

Planning schemes	TC (€/a)	CONL (kW)	Cost of complementarity enhancement (€/ (a·kW))
1	670480.82	545.41	223.6
7	683792.77	485.89	
14	719851.2	387.25	839.9
25	776199.45	320.16	

The annual investment and maintenance costs generally increase as the value of the complementarity function decreases (the complementarity of IES increases). In the planning scheme with stronger complementarity, the

annualized total cost is larger and the devices require more expenses.

In a different range of complementarity, the relationship between CONL and TC indicators is not linear. When complementarity is poor, less cost can achieve more significant complementarity enlargement. Table 1 shows the relationship between CONL change and TC change between some planning schemes with different complementary. Measured by the indicators proposed in this paper, the annual total costs will increase by 15.8% when the system complementarity is enhanced by 41.3%. From planning schemes 1 to 7, reducing CONL by 1kW costs 223.6€ per year. From planning schemes 14 to 25, the annualized total costs of reducing CONL by 1kW are 839.9€, which is 3.75 times more than the former.

V. Conclusions

The two-stage planning method of IES considering economy and complementarity proposed in this paper generates various planning schemes with different economy and complementarity, which can be used to guide the optimization of device configuration. In the IES long-term planning optimization, economy and complementarity are two contradictory indicators normally. The improvement of system complementarity will lead to economic decline. However, it is worth noting that developing appropriate strategies can achieve greater complementarity with lower investment costs since the marginal cost of enhancing complementarity is low in the initial stage. A case study of regional integrated energy systems for industrial users shows that the annual total costs will increase by 15.8% when the system complementarity is enhanced by 41.3%. Meanwhile, the yearly cost of complementarity enhancement is just 223.6 €/kW when the complementarity index is about 500kW.

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